SET 63 • HSS-ID.B.6.a

BACKWARD 1

Describe patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

2402

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is an outlier in a scatter plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

2403

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A point far from the overall pattern of the others

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Points fall from left to right in a curved pattern.

Which describes the association?

A. Positive and nonlinear

B. Positive and linear

C. Negative and nonlinear

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

2404

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Negative and nonlinear

GRADING NOTE Falling means negative; curved means nonlinear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

True or false?

A scatter plot shows the relationship between two quantitative variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

2405

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point is a pair of values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

True or false?

If the points show no pattern, there is a strong association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

2406

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pattern means little or no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 2

Write linear and exponential functions through given points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function through (0, 100) and (1, 130).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

2408

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.3)^x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, 2) and (4, 26).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

2409

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

Which exponential function $y = ab^x$ $(b > 0)$ passes through $(0, 50)$ and $(2, 12.5)$?

A. $y = 50(0.25)^x$

B. $y = 12.5(2)^x$

C. $y = 50(0.5)^x$

D. $y = 50 - 18.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

2410

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 50(0.5)^x$

GRADING NOTE $a = 50$ and $50b^2 = 12.5$, so $b = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 3(2)^x$ passes through (2, 12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

2411

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

True or false?

$y = 5(1.2)^x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

2412

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 1

Use logarithms with an exponential model fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$P(t) = 100(1.3)^t$.

When does P reach 1,000? Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

2414

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.8

GRADING NOTE Form required: rounded to the nearest tenth. t = log 10 ÷ log 1.3 ≈ 8.78.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2(1.5)^x = 30$. Round to the nearest hundredth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

2415

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.68

GRADING NOTE Form required: rounded to the nearest hundredth. $1.5^x = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

Which is the exact solution of $100(1.3)^t = 500$? (log means base 10.)

A. t = log 1.3 ÷ log 5

B. t = 500 ÷ 130

C. t = log 5 ÷ log 1.3

D. t = 5 ÷ 1.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

2416

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = log 5 ÷ log 1.3

GRADING NOTE $1.3^t = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

True or false?

If $1.3^t = 1.69$, then t = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

2417

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.3^2 = 1.69$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $10^t = 50$, then $t = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

2418

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = \log 50 \approx 1.70$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 2

Fit a sinusoidal model to periodic data • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Monthly average temperatures peak at 30 °C and bottom out at 10 °C.

What amplitude and midline fit a sinusoidal model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

2420

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Amplitude 10; midline y = 20

GRADING NOTE Form required: both. (30 − 10) ÷ 2 and (30 + 10) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What period fits monthly temperatures that repeat every year?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

2421

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
12 months

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

Daylight hours vary from 9 to 15 over a year.

Which model fits?

A. A linear model

B. A sinusoidal model with amplitude 3 and period 12 months

C. A quadratic model with vertex at month 12

D. An exponential model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

2422

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sinusoidal model with amplitude 3 and period 12 months

GRADING NOTE The pattern repeats each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

True or false?

A repeating pattern in data suggests a sinusoidal model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

2423

QUESTION ID
S63-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Sinusoids are periodic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

True or false?

Data that vary between 9 and 15 have amplitude 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

2424

QUESTION ID
S63-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is half the distance: 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Construct scatter plots from paired data and describe how the variables are related

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Make and describe scatter plots	**8.SP.A.1**
BACKWARD 2	Plot points and find distances on the coordinate plane	**6.NS.C.8**
FORWARD 1	Choose an exponential model from a scatter plot and solve it with…	**HSF-LE.A.4**
FORWARD 2	Evaluate claims based on scatter plots	**HSS-IC.B.6**

SET 83 • HSS-ID.B.6.a

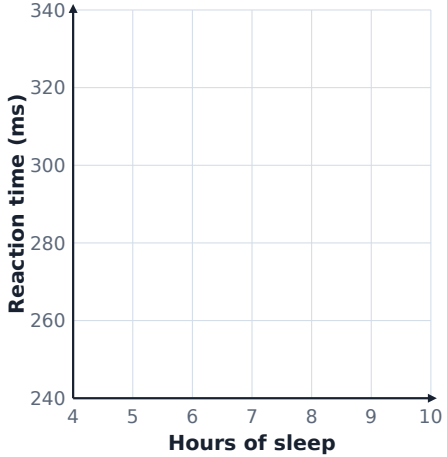
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

Make a scatter plot of hours of sleep (x) and reaction time in milliseconds (y):

(5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

2427

QUESTION ID

S83-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Six points plotted at the given pairs, falling from left to right

GRADING NOTE Any correct scatter plot with all six points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the association between hours of sleep and reaction time in the data (5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255): its direction, form, and strength.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

2428

QUESTION ID
S83-M-Q2

TYPE
Short answer

CORRECT ANSWER
Negative, roughly linear, and strong: more sleep goes with shorter reaction times.

GRADING NOTE Form required: direction, form, and strength.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

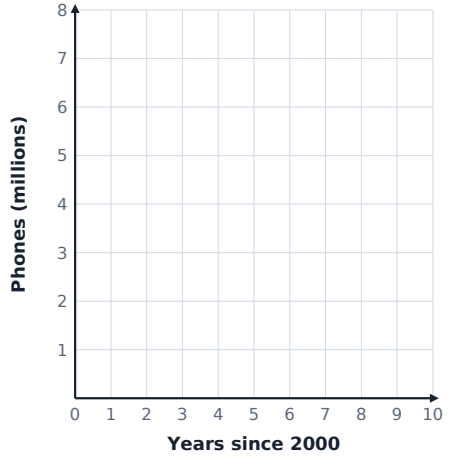
ANSWER KEY Reduced copy of the question page, with its answer

Make a scatter plot of years since 2000 (x) and phones in millions (y):

(0, 1), (2, 1.5), (4, 2.3), (6, 3.4), (8, 5.1), (10, 7.6)

Which type of model does the plot suggest?

Model: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

2429

QUESTION ID

S83-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Six points rising from left to right and curving upward

Written: Exponential: the points curve upward, growing by about 1.5 times every 2 years.

GRADING NOTE Form required: the model type with a reason. Both the plot and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

When you make a scatter plot to see whether one variable helps explain another, which variable usually goes on the horizontal axis?

A. Whichever variable has larger numbers

B. Whichever variable has fewer values

C. The explanatory variable (the one that may explain the other)

D. The response variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

2430

QUESTION ID

S83-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The explanatory variable (the one that may explain the other)

GRADING NOTE By convention the explanatory variable is x and the response variable is y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

True or false?

Both axes of a scatter plot must start at 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

2431

QUESTION ID

S83-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Axes can start near the data, as long as the scales are labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

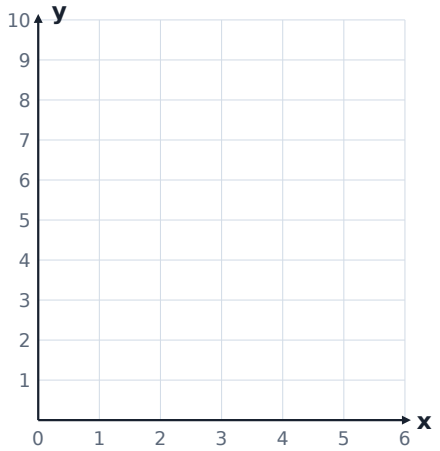
SET 83 • HSS-ID.B.6.a

BACKWARD 1

Make and describe scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Plot (1, 2), (2, 3), (3, 5), (4, 6), and (5, 8) as a scatter plot.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

2433

QUESTION ID

S83-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Five points rising from left to right

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right but curve upward. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

2434

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive and nonlinear

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes a scatter plot whose points show no trend?

A. Linear association

B. Strong positive association

C. Strong negative association

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

2435

QUESTION ID

S83-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No association

GRADING NOTE No trend means no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

True or false?

Each point in a scatter plot represents one pair of values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

2436

QUESTION ID

S83-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One point per pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A cluster in a scatter plot is the same thing as an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

2437

QUESTION ID

S83-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A cluster is a group of points; an outlier is a single point far from the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

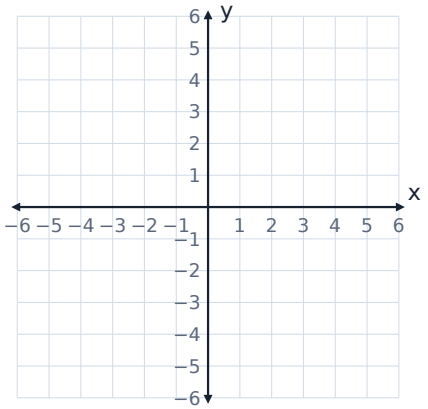
SET 83 • HSS-ID.B.6.a

BACKWARD 2

Plot points and find distances on the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Plot the points (2, 5), (−3, 1), and (0, −4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

2439

QUESTION ID

S83-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Three points at (2, 5), (−3, 1), and (0, −4)

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (−3, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

2440

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

GRADING NOTE Any equivalent form. |5 − (−3)| = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

Which point lies on the y-axis?

A. (1, −1)

B. (0, −4)

C. (−4, 0)

D. (4, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

2441

QUESTION ID

S83-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, −4)

GRADING NOTE Points on the y-axis have x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

True or false?

The point (−2, 6) is in Quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

2442

QUESTION ID

S83-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Quadrant II has $x < 0$ and $y > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The distance between (1, −3) and (1, 4) is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

2443

QUESTION ID

S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $|4 - (-3)| = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 1

Choose an exponential model from a scatter plot and solve it with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A model for the phone data is $y = 1(1.22)^x$, where x is years since 2000 and y is millions of phones. When does it predict 20 million? Round to the nearest tenth. (log means base 10.)

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

2445

QUESTION ID
S83-F1-Q1

TYPE
Short answer

CORRECT ANSWER
15.1 years after 2000

GRADING NOTE Form required: rounded to the nearest tenth. x = log 20 ÷ log 1.22 ≈ 15.07.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using $y = 1(1.22)^x$, how many years does it take the number of phones to double? Round to the nearest tenth. (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

2446

QUESTION ID
S83-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3.5

GRADING NOTE Form required: rounded to the nearest tenth. log 2 ÷ log 1.22 ≈ 3.49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

Solve $4(1.25)^x = 20$. (log means base 10.)

A. $x = \log 5 \div \log 1.25 \approx 7.21$

B. $x = 5 \div 1.25$

C. $x = 4$

D. $x = \log 20 \div \log 1.25 \approx 13.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

2447

QUESTION ID
S83-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $x = \log 5 \div \log 1.25 \approx 7.21$

GRADING NOTE $1.25^x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

True or false?

If $y = a \cdot b^x$ with $a > 0$ and $b > 1$, then y doubles every log 2 ÷ log b units of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

2448

QUESTION ID

S83-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $b^d = 2$ gives $d = \log 2 \div \log b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The model $y = 1(1.22)^x$ predicts 0 phones in 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

2449

QUESTION ID
S83-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At x = 0, y = 1 (million).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 2

Evaluate claims based on scatter plots • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A website plots shoe size against reading score for students in Grades 1 to 8, finds a strong positive association, and claims that bigger feet improve reading. What is wrong?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

2451

QUESTION ID
S83-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Age (grade) is a lurking variable: older students have bigger feet and read better. Observational data do not show cause.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A scatter plot of 6 students from one class shows a strong trend. Why should you be careful applying it to all students?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

2452

QUESTION ID
S83-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The sample is small and was not randomly selected, so it may not represent all students.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

Which detail would make a claim based on a scatter plot more trustworthy?

A. Both axes start at 0.

B. The plot uses bright colors.

C. There are only 5 points.

D. The data come from a large random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

2453

QUESTION ID
S83-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. The data come from a large random sample.

GRADING NOTE Large random samples represent the population better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q3**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

True or false?

Extrapolating a trend far beyond the plotted data can give misleading predictions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

2454

QUESTION ID

S83-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The pattern may not continue outside the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

True or false?

If a scatter plot shows a strong trend, one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

2455

QUESTION ID
S83-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Association does not show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Assess the fit of a function by plotting and analyzing residuals

MAIN	Algebra I main question	**HSS-ID.B.6.b**
BACKWARD 1	Use a fitted function to make predictions	**HSS-ID.B.6.a**
BACKWARD 2	Subtract rational numbers	**7.NS.A.1.c**
FORWARD 1	Use logarithms to solve exponential models fitted to data	**HSF-LE.A.4**
FORWARD 2	Evaluate reports that use fitted models	**HSS-IC.B.6**

SET 64 • HSS-ID.B.6.b

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

ANSWER KEY Reduced copy of the question page, with its answer

The model $\hat{y} = 2.2x + 0.8$ is fitted to the data (1, 3), (2, 5), (3, 8), (4, 9), (5, 12).

Find the residual for each point.

Residuals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

2458

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

0, −0.2, 0.6, −0.6, 0.2

GRADING NOTE Form required: all five, in order. Residual = actual − predicted; the predictions are 3, 5.2, 7.4, 9.6, 11.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

For that model, what does the residual −0.6 at x = 4 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

2459

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

The actual y-value is 0.6 less than the model predicts.

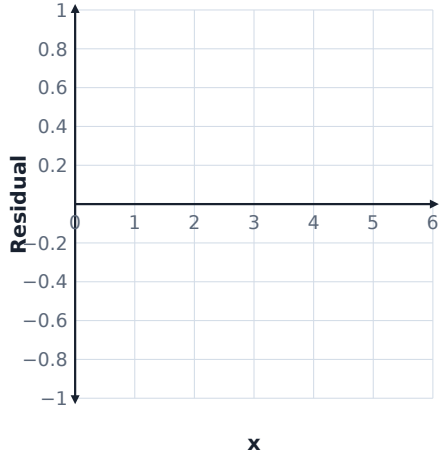
GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Make a residual plot for the residuals 0, −0.2, 0.6, −0.6, 0.2 at x = 1, 2, 3, 4, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

2460

QUESTION ID

S64-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The points (1, 0), (2, −0.2), (3, 0.6), (4, −0.6), and (5, 0.2), scattered above and below 0 with no pattern

GRADING NOTE Any correct plot. The lack of a pattern supports a linear form; the small size of the residuals (all within 0.6) shows the errors are small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

2460

A residual plot shows a clear U-shaped pattern.

What does that suggest?

A. The correlation is exactly 1.

B. A linear model is not a good fit; a curved model may fit better.

C. The linear model fits well.

D. There are no residuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

2461

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A linear model is not a good fit; a curved model may fit better.

GRADING NOTE A pattern in the residuals means the model misses part of the relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

True or false?

A good linear fit has residuals scattered randomly around 0, with no pattern.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

2462

QUESTION ID
S64-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The model then captures the pattern of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 1

Use a fitted function to make predictions • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The model is $\hat{y} = 2.2x + 0.8$.

Predict y when x = 6.

$\hat{y}$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

2464

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 13.2 + 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The model is $y = 3x^2 - 2$.

Predict y when x = 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

2465

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

46

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The model is $y = 50(0.9)^x$.

Predict y when x = 2.

A. 50

B. 40.5

C. 45

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

2466

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40.5

GRADING NOTE 50(0.81) = 40.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

True or false?

A predicted value comes from the model, not from the data table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

2467

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the model's output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

True or false?

If a model predicts 14 and the actual value is 14, the residual is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

2468

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The residual is $14 - 14 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 8 − 7.4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

2470

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 9 − 9.6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

2471

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 5 − 5.2.

A. −10.2

B. 10.2

C. 0.2

D. −0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

2472

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.2

GRADING NOTE 5.2 is greater than 5, so the difference is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$3 - (-2) = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

2473

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtracting −2 adds 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-4 - 6 = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

2474

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4 - 6 = -10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 1

Use logarithms to solve exponential models fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A model fitted to bacteria counts is $N = 150(1.4)^t$, with t in hours. When does N reach 1,000? Give the exact expression and the value to the nearest tenth of an hour. (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

2476

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

t

= log(1,000 ÷ 150) ÷ log 1.4 ≈ 5.6 hours

GRADING NOTE Form required: the exact expression (any equivalent, such as log(20/3) ÷ log 1.4) and the rounded value. $1.4^t = 6.67$, $t \approx 5.64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line fitted to the logarithms of data is log y = 0.3x + 1 (log means base 10).

Write y as an exponential function of x.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

2477

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$10 \cdot 10^{0.3x}$

GRADING NOTE Any equivalent exact form, such as $10^{0.3x+1}$ or $10(10^{0.3})^x$. A rounded base such as $10(2)^x$ is not exact: $10^{0.3} \approx 1.995$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fitted model is $y = 20e^{0.08x}$. For which x is y = 60? (ln means base e.)

A. $x = 3 \div 0.08 = 37.5$

B. $x = \ln 60 \div 0.08 \approx 51.2$

C. $x = 0.08 \ln 3 \approx 0.09$

D. $x = \ln 3 \div 0.08 \approx 13.7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

2478

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \ln 3 \div 0.08 \approx 13.7$

GRADING NOTE $e^{0.08x} = 3$, so $0.08x = \ln 3 \approx 1.0986$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For the model $y = 5(1.5)^x$, the solution of $5(1.5)^x = 50$ is $x = \log 10 \div \log 1.5 \approx 5.68$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

2479

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.5^x = 10$, so $x = \log 10 \div \log 1.5 = 1 \div 0.1761 \approx 5.68$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

The doubling time of $N = 150(1.4)^t$ is log 2 ÷ log 1.4 ≈ 0.49 hour. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

2480

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE log 2 ÷ log 1.4 ≈ 0.301 ÷ 0.146 ≈ 2.06 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 2

Evaluate reports that use fitted models • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A report fits a line to the heights of children aged 5 to 15 and uses it to predict the height of a 40-year-old.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

2482

QUESTION ID
S64-F2-Q1

TYPE
Short answer

CORRECT ANSWER

It extrapolates far beyond the data; people stop growing, so the linear pattern does not continue.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report finds a strong correlation between ice cream sales and drownings and concludes that ice cream causes drowning.

Do these observational data establish that conclusion? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

2483

QUESTION ID
S64-F2-Q2

TYPE
Short answer

CORRECT ANSWER

No. A correlation in observational data does not establish causation; a lurking variable, such as hot weather, could increase both.

GRADING NOTE Any equivalent explanation that association alone does not establish causation. Naming a possible lurking variable is accepted but not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

A residual plot shows a strong curve, but a report says the linear model is excellent.

What should you conclude?

A. The report is correct.

B. The data must be wrong.

C. The residuals are all 0.

D. The linear model may not be appropriate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

2484

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The linear model may not be appropriate.

GRADING NOTE A curved residual pattern shows a poor linear fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

True or false?

A report should state how its data were collected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

2485

QUESTION ID
S64-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The method decides what conclusions are justified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

True or false?

A model with a high correlation coefficient is always the right type of model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

2486

QUESTION ID
S64-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Curved data can have a high r for a line and still be fit better by a curve.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear function to a scatter plot that suggests a linear association

MAIN	Algebra I main question	**HSS-ID.B.6.c**
BACKWARD 1	Judge a line of fit informally	**8.SP.A.2**
BACKWARD 2	Write a line through two points	**HSF-LE.A.2**
FORWARD 1	Evaluate conclusions drawn from a line of fit	**HSS-IC.B.6**
FORWARD 2	Use a randomized experiment and simulation to test whether mo…	**HSS-IC.B.5**

SET 65 • HSS-ID.B.6.c

MAIN

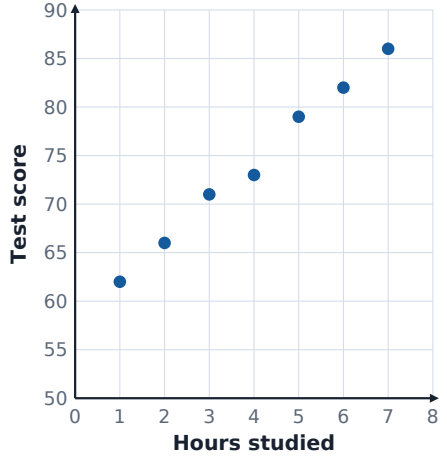
Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows hours studied and test scores.

Draw a line of fit and write its equation.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

2489

QUESTION ID

S65-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line that follows the trend of the points closely (balancing points above and below is a helpful check)

Written: About y = 4x + 58

GRADING NOTE Any equation close to the least-squares line $\hat{y} = 4x + 58.14$ (slope about 3.5 to 4.5). The equation should match the student's line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line for the data is $\hat{y} = 4x + 58.14$.

Predict the score for 4.5 hours, to the nearest whole number.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

2490

QUESTION ID

S65-M-Q2

TYPE

Short answer

CORRECT ANSWER

76

GRADING NOTE Form required: rounded to the nearest whole number. 18 + 58.14 = 76.14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through the data points (1, 62) and (7, 86), to use as a line of fit.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

2491

QUESTION ID

S65-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 4x + 58

GRADING NOTE Any equivalent form. The slope is 24 ÷ 6 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line fits the points (0, 1), (1, 3), (2, 5), (3, 7) exactly?

A. y = 3x + 1

B. y = 2x + 1

C. y = 2x

D. y = x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

2492

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 1

GRADING NOTE y increases by 2 for each 1 in x, starting at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

True or false?

A line of fit should pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

2493

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It should be close to the points overall, not through all of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 1

Judge a line of fit informally • Grade 8 • Statistics & Probability • 8.SP.A.2

Is a straight line a good model for points that curve strongly upward? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

2495

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The points do not follow a straight pattern.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How can you judge whether a line of fit is good?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

2496

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The points are close to the line and follow its trend, with no pattern of points far above or below.

GRADING NOTE Any equivalent description. Balancing points above and below is a helpful check, not a definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

2496

ANSWER KEY Reduced copy of the question page, with its answer

Which is the best sign of a good line of fit?

A. All of the points are below it.

B. All of the points are above it.

C. The points are close to the line and follow its trend.

D. Exactly two points are on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

2497

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The points are close to the line and follow its trend.

GRADING NOTE A good line stays close to the points; balancing points above and below is a helpful check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

True or false?

Lines of fit are used for data with a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

2498

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Straight-line patterns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line of fit must pass through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

2499

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its y-intercept can be any value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 2

Write a line through two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (2, 66) and (6, 82).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

2501

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 58

GRADING NOTE Any equivalent form. The slope is 16 ÷ 4 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (0, 7) and (3, 13).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

2502

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line passes through (0, 10) and (5, 0)?

A. $y = 2x + 10$

B. $y = 10x - 2$

C. $y = -2x + 10$

D. $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

2503

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -2x + 10$

GRADING NOTE The slope is $-10 \div 5 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The line through (0, 3) with slope 4 is y = 4x + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

2504

QUESTION ID
S65-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE y = mx + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The line through (2, 2) and (4, 8) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

2505

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $6 \div 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 1

Evaluate conclusions drawn from a line of fit • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

In a study of 7 volunteer students, each extra hour of study went with about 4 more points.

Can a school claim this holds for all of its students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

2507

QUESTION ID

S65-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Not with confidence: 7 volunteers are a small sample that was not randomly selected, and an observational study shows association, not cause.

GRADING NOTE Any equivalent explanation that names the sampling problem (small, not random) or the association-causation limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study of 7 students who studied from 1 to 7 hours found the line $\hat{y} = 4x + 58$, where x is hours of study and $\hat{y}$ is the test score out of 100.

A report uses this line to predict a score of 138 for 20 hours of study. Why is that unreasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

2508

QUESTION ID
S65-F1-Q2

TYPE
Short answer

CORRECT ANSWER
20 hours is far outside the data (1 to 7 hours), and the score cannot exceed 100.

GRADING NOTE Any equivalent explanation that names the extrapolation or the 100-point maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which detail makes a regression conclusion more trustworthy?

A. The prediction is far outside the data.

B. The line was drawn by hand.

C. The data came from a large random sample.

D. The sample was 5 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

2509

QUESTION ID
S65-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. The data came from a large random sample.

GRADING NOTE Random, large samples represent the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

True or false?

Observational data alone cannot prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

2510

QUESTION ID
S65-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Lurking variables may explain the association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

True or false?

A line of fit from 7 volunteer students proves that extra study raises the score of every student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

2511

QUESTION ID
S65-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It describes an association in that small, nonrandom sample; it does not prove an effect for everyone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 2

Use a randomized experiment and simulation to test whether more study raises scores • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

To test whether extra study raises scores, 40 volunteers were randomly assigned: 20 studied 3 extra hours and 20 did not. The extra-study group averaged 5 points higher. In 1,000 random regroupings of the 40 scores, a difference of 5 points or more (first group minus second) happened 18 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2513

QUESTION ID
S65-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 18 ÷ 1,000 = 1.8%, which is below 5%, so chance alone is unlikely to explain the difference; the extra study likely raised scores for volunteers like these.

GRADING NOTE Form required: the decision with the simulated proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a randomized experiment, a tutoring group averaged 6 points more than a control group. Random regroupings of the scores gave differences of 6 points or more 30% of the time.

Using a 5% threshold, what is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2514

QUESTION ID
S65-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The difference is not significant: 30% is above 5%, so it could easily be due to chance.

GRADING NOTE Form required: the conclusion with the comparison to 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the purpose of random assignment in an experiment?

A. To balance other variables between the groups, on average

B. To make the sample smaller

C. To guarantee a significant result

D. To remove the need for a control group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2515

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. To balance other variables between the groups, on average

GRADING NOTE Random assignment tends to balance other variables, so a large difference can be credited to the treatment. It does not make the groups identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

True or false?

A result is statistically significant when a difference that large would rarely happen by chance alone, using a threshold chosen in advance such as 5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2516

QUESTION ID
S65-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is the meaning of significance in a randomization test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

True or false?

Using a 5% threshold, a difference that happens in 40% of random regroupings is statistically significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2517

QUESTION ID
S65-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 40% is far above 5%, so such a difference is common by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

Interpret the slope and the intercept of a linear model in the context of the data

MAIN	Algebra I main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**
BACKWARD 2	Use a linear model to solve problems	**8.SP.A.3**
FORWARD 1	Interpret the parameters of continuous exponential models	**HSF-LE.A.4**
FORWARD 2	Understand a fitted slope as an estimate of a population value	**HSS-IC.A.1**

SET 66 • HSS-ID.C.7

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 4x + 58$ predicts a test score from x hours of study.

Interpret the slope and the intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

2520

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: each additional hour of study goes with about 4 more points (predicted). Intercept: the predicted score for 0 hours is 58.

GRADING NOTE Form required: both interpretations, in context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for used cars is $\hat{y} = 22{,}000 - 1{,}800x$, where x is the age in years and $\hat{y}$ is the price in dollars.

Interpret $-1{,}800$ and $22{,}000$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

2521

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

The predicted price drops about $1,800 for each year of age; the predicted price of a new (0-year-old) car is $22,000.

GRADING NOTE Form required: both interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 0.9x - 20$ predicts an adult's weight in pounds from height in inches.

Does the intercept make sense in context? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

2522

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. It predicts −20 pounds at a height of 0 inches, which is impossible and far outside the data.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

$\hat{y} = 2.5x + 10$ predicts the cost in dollars of a pizza with x toppings.

What does 2.5 mean?

A. Each additional topping adds about $2.50 to the predicted cost.

B. A pizza has 2.5 toppings.

C. A pizza costs $2.50.

D. A plain pizza costs $2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

2523

QUESTION ID

S66-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Each additional topping adds about $2.50 to the predicted cost.

GRADING NOTE The slope is the change in predicted cost per topping.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

True or false?

The slope of a fitted line gives the exact change in y for every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

2524

QUESTION ID

S66-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives the predicted (average) change; individuals vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 1

Interpret the parameters of linear and exponential functions • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A technician charges C = 50 + 20h dollars for h hours.

What does 20 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

2526

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$20 for each hour of work

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 3{,}000(0.9)^t$ is a population after t years.

What does 0.9 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

2527

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The population decreases 10% each year.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$d = 300 - 60t$ is a driver's distance from home, in miles, after t hours.

What does 300 mean?

A. The starting distance from home

B. The speed

C. The distance driven each hour

D. The time to arrive home

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

2528

QUESTION ID
S66-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. The starting distance from home

GRADING NOTE At $t = 0$, $d = 300$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

True or false?

In y = 1.5x + 4, increasing x by 2 increases y by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

2529

QUESTION ID
S66-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 1.5 × 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $y = 8(1.25)^x$, y increases by 1.25 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

2530

QUESTION ID
S66-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE y is multiplied by 1.25 for each unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

2530

SET 66 • HSS-ID.C.7

BACKWARD 2

Use a linear model to solve problems • Grade 8 • Statistics & Probability • 8.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

y = 2x + 5 models a plant's height in cm after x weeks.

Predict the height at 6 weeks.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

2532

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

17 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using y = 2x + 5, after how many weeks is the plant 21 cm tall?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

2533

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2x + 5 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 2x + 5$ models a plant's height in cm after x weeks.

In this model, what does 5 mean?

A. The height at week 0

B. The weeks needed to grow 1 cm

C. The final height

D. The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

2534

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The height at week 0

GRADING NOTE $y = 5$ when $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope of a linear model is a rate of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

2535

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the change in y per unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 2x + 5, the plant grows 5 cm each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

2536

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It grows 2 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 1

Interpret the parameters of continuous exponential models • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A(t) = 500e^{0.04t}$.

What does 0.04 mean? When does A double, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

2538

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A continuous growth rate of 4% per unit of t; t ≈ 17.3

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$N(t) = 1{,}000e^{-0.1t}$.

What does −0.1 mean? When is N = 500, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

2539

QUESTION ID
S66-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Continuous decay at 10% per unit of t; t ≈ 6.9

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{2x} = 9$ exactly. (ln means base e.)

A. $x = \ln 3$

B. $x = 4.5$

C. $x = e^{4.5}$

D. $x = \ln 9 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

2540

QUESTION ID
S66-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. x = ln 3

GRADING NOTE $2x = \ln 9 = 2 \ln 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\ln(e^{0.04t}) = 0.04t$ (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

2541

QUESTION ID

S66-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ln and e^x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $e^k = 1$, then $k = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

2542

QUESTION ID

S66-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $e^0 = 1$, so $k = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 2

Understand a fitted slope as an estimate of a population value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 50 used cars of one model gives $\hat{y} = 22{,}000 - 1{,}800x$.

What population value does $-1{,}800$ estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

2544

QUESTION ID
S66-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The average yearly drop in price for all used cars of that model.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Would another random sample of 50 cars give exactly the same slope? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

2545

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Probably not: different samples give different statistics (sampling variability).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which of these is a parameter?

A. The mean price of the 50 sampled cars

B. One car's price

C. The slope from the sample

D. The mean price of all used cars of this model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

2546

QUESTION ID

S66-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The mean price of all used cars of this model

GRADING NOTE A parameter describes the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

True or false?

Larger random samples tend to give statistics closer to the parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

2547

QUESTION ID
S66-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Their sampling variability is smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

True or false?

A statistic is a number that describes a whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

2548

QUESTION ID
S66-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

Compute (using technology) and interpret the correlation coefficient of a linear fit

MAIN	Algebra I main question	**HSS-ID.C.8**
BACKWARD 1	Describe the direction of an association	**8.SP.A.1**
BACKWARD 2	Fit and use a line for linear data	**HSS-ID.B.6.c**
FORWARD 1	Use a randomized experiment and simulation to test whether an a…	**HSS-IC.B.5**
FORWARD 2	Evaluate claims based on a correlation	**HSS-IC.B.6**

SET 67 • HSS-ID.C.8

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use technology to find the correlation coefficient r for the data

(1, 9), (2, 8), (3, 8), (4, 6), (5, 5), (6, 5), (7, 3), (8, 2),

to the nearest hundredth. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

2551

QUESTION ID

S67-M-Q1

TYPE

Short answer

CORRECT ANSWER

r ≈ −0.98: a strong negative linear association.

GRADING NOTE Form required: r to the nearest hundredth and a description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$r = -0.85$ for hours of TV and test score.

Interpret this value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

2552

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

A strong negative linear association: more TV goes with lower scores.

GRADING NOTE Any equivalent interpretation of the direction, strength, and form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value of r shows the strongest linear relationship?

A. 0.10

B. −0.92

C. −0.40

D. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

2553

QUESTION ID

S67-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −0.92

GRADING NOTE Strength depends on |r|, and |−0.92| is the largest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Can a correlation coefficient equal 1.3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

2554

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. r is always between −1 and 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

True or false?

r = 0 means there is no relationship of any kind between the variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

2555

QUESTION ID
S67-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means no linear relationship; there could be a curved one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 1

Describe the direction of an association • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, fairly close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

2557

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points for hours of sleep and number of errors fall from left to right.

Is the association positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

2558

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of variables likely has a positive association?

A. Height and arm span

B. Outdoor temperature and heating cost

C. A car's age and its value

D. Speed and travel time for a fixed trip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

2559

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Height and arm span

GRADING NOTE Taller people tend to have longer arm spans.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a negative association, as one variable increases, the other tends to decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

2560

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the meaning of negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Clusters in a scatter plot mean the data are linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

2561

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Clusters describe grouping, not form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 2

Fit and use a line for linear data • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Write a line of fit through (2, 5) and (8, 14).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

2563

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5x + 2

GRADING NOTE Any equivalent form. The slope is 9 ÷ 6 = 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line is $\hat{y} = 1.5x + 2$.

Predict y when $x = 6$.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

2564

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What does the least-squares line minimize?

A. The y-intercept

B. The sum of the squared residuals

C. The number of points

D. The slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

2565

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The sum of the squared residuals

GRADING NOTE That is what "least squares" means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

2566

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of the least-squares line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

True or false?

Two people drawing lines of fit by eye always get the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

2567

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines drawn by eye vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 1

Use a randomized experiment and simulation to test whether an association is causal • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Observational data show r = −0.6 between evening screen time and hours of sleep. To test cause, 50 volunteers were randomly assigned to no screens after 8 p.m. or to their usual habits. The no-screen group slept 0.5 hour more on average. In 1,000 random regroupings, a difference of at least 0.5 hour (no-screen minus usual) happened 2 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

2569

QUESTION ID
S67-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 2 ÷ 1,000 = 0.2%, which is below 5%, so chance alone is very unlikely to explain it; cutting evening screens likely increased sleep for volunteers like these.

GRADING NOTE Form required: the decision with the simulated proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can this randomized experiment, but not the observational correlation alone, support the claim that evening screens reduce sleep?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

2570

QUESTION ID

S67-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Random assignment tends to balance other variables (such as stress or age) between the groups, so a significant difference can be credited to the treatment; in observational data a lurking variable could explain the correlation.

GRADING NOTE Any equivalent explanation that contrasts random assignment with possible lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a randomization test, the observed difference was matched or exceeded in 350 of 1,000 regroupings.

Using a 5% threshold, what is the conclusion?

A. The treatment definitely has no effect.

B. The difference is not significant; chance could easily explain it.

C. The difference is significant.

D. The experiment proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

2571

QUESTION ID

S67-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The difference is not significant; chance could easily explain it.

GRADING NOTE 350 ÷ 1,000 = 35%, far above 5%. A non-significant result does not prove there is no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

True or false?

In a randomization test, the regroupings show how large the difference between groups tends to be when the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

2572

QUESTION ID
S67-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Regrouping the same results at random imitates assignment with no treatment effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

True or false?

If an experiment's difference is not significant, that proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

2573

QUESTION ID

S67-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means the data do not show an effect beyond chance; a small effect could still exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 2

Evaluate claims based on a correlation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A news story says, "r = 0.6, so 60% of the change in sales is caused by advertising."

Name two errors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2575

QUESTION ID
S67-F2-Q1

TYPE
Short answer

CORRECT ANSWER
r is not a percent of change ($r^2 \approx 0.36$ describes the share of variation explained), and correlation does not show causation.

GRADING NOTE Form required: two errors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study of 8 cities finds r = 0.7 between the number of parks and happiness scores.

Why should you be cautious?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2576

QUESTION ID

S67-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The sample is small, the study is observational, and other variables (such as income) could explain both.

GRADING NOTE Any two valid reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement about r = −0.9 is correct?

A. There is a strong negative linear association.

B. The slope of the line is −0.9.

C. 90% of the points are on the line.

D. An increase in x causes y to fall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2577

QUESTION ID

S67-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. There is a strong negative linear association.

GRADING NOTE r measures the direction and strength of a linear association only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

True or false?

A correlation can be strong even when the relationship is not causal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2578

QUESTION ID
S67-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A lurking variable can create it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

If r = 0.8, the slope of the least-squares line is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2579

QUESTION ID
S67-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The slope depends on the units and spreads of x and y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-ID.C.9 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Distinguish between correlation and causation

MAIN	Algebra I main question	**HSS-ID.C.9**
BACKWARD 1	Interpret the correlation coefficient	**HSS-ID.C.8**
BACKWARD 2	Use random samples to make valid generalizations	**7.SP.A.1**
FORWARD 1	Distinguish surveys, experiments, and observational studies	**HSS-IC.B.3**
FORWARD 2	Use random methods to make fair decisions — nearest related	**HSS-MD.B.6**

SET 68 • HSS-ID.C.9

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Data show that cities with more firefighters have more fire damage.

Do these data establish that hiring firefighters causes damage? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

2582

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

No. The data show an association only. A lurking variable, such as the size of the city or of its fires, could explain both.

GRADING NOTE Any equivalent explanation that association does not establish causation. A plausible lurking variable is accepted but not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

Ice cream sales and sunburn rates are strongly correlated.

Name a lurking variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

2583

QUESTION ID

S68-M-Q2

TYPE

Short answer

CORRECT ANSWER

Hot, sunny weather

GRADING NOTE Any reasonable lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How could you test whether a new study app causes higher test scores?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

2584

QUESTION ID
S68-M-Q3

TYPE
Short answer

CORRECT ANSWER
Run a randomized experiment: randomly assign students to use the app or not, then compare the groups' scores.

GRADING NOTE Any design that uses random assignment and a comparison group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

Which study design can show that one variable causes changes in another?

A. A survey of volunteers

B. A scatter plot of existing data

C. An observational study

D. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

2585

QUESTION ID

S68-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A randomized experiment

GRADING NOTE Random assignment balances other variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

True or false?

A strong correlation between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

2586

QUESTION ID

S68-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Correlation alone never proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 1

Interpret the correlation coefficient • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

ANSWER KEY Reduced copy of the question page, with its answer

r = 0.88.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

2588

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which shows the stronger linear association, r = −0.7 or r = 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

2589

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

r = −0.7

GRADING NOTE Any equivalent form. |−0.7| > |0.5|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is r for points that lie exactly on a line with negative slope?

A. 1

B. 0

C. −0.5

D. −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

2590

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −1

GRADING NOTE A perfect negative linear relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

The correlation coefficient has no units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

2591

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is computed from standardized values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

Switching the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

2592

QUESTION ID
S68-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r is the same either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 2

Use random samples to make valid generalizations • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why is a random sample more likely to represent a population than a convenience sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

2594

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Every member has the same chance of being chosen, so the method does not favor any group (no selection bias), although a random sample can still differ from the population by chance.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

A survey at a gym asks about the exercise habits of all town residents.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

2595

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

People at a gym probably exercise more than other residents, so the sample is biased.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a random sample of a school's students?

A. Drawing 40 names from a list of all students

B. Asking the first 40 students in line

C. Asking your friends

D. Asking one math class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

2596

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Drawing 40 names from a list of all students

GRADING NOTE Only the drawing gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

Generalizations about a population are valid only if the sample represents the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

2597

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is why random sampling matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

2598

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 1

Distinguish surveys, experiments, and observational studies • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Researchers record students' screen time and grades without changing anything.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

2600

QUESTION ID
S68-F1-Q1

TYPE
Short answer

CORRECT ANSWER
An observational study

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Researchers randomly assign plants to receive fertilizer or no fertilizer.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

2601

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

An experiment

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the main purpose of a survey?

A. To test whether a treatment causes an effect

B. To assign treatments to subjects

C. To control lurking variables

D. To estimate a characteristic of a population from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

2602

QUESTION ID
S68-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. To estimate a characteristic of a population from a sample

GRADING NOTE A survey collects information without assigning treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

True or false?

Random selection lets you generalize to a population; random assignment lets you conclude cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

2603

QUESTION ID
S68-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two kinds of randomization do different jobs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

True or false?

In an observational study, the researchers decide which treatment each subject gets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

2604

QUESTION ID
S68-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They only observe.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 2

Use random methods to make fair decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Random assignment, which is needed to show causation, is a fair random method.

ANSWER KEY Reduced copy of the question page, with its answer

Describe a fair way to split 20 volunteers into two groups of 10 using random numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

2606

QUESTION ID
S68-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Number the volunteers 1 to 20, randomly draw 10 different numbers for group A, and put the rest in group B.

GRADING NOTE Any method that gives every volunteer the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A coach picks a captain by drawing one name from a hat with 12 names.

What is the probability that a given player is chosen?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

2607

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Which method is fair for deciding who goes first among 4 players?

A. The tallest goes first

B. Each rolls a number cube; ties roll again; the highest goes first

C. The coach's favorite goes first

D. Whoever asks first goes first

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

2608

QUESTION ID
S68-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. Each rolls a number cube; ties roll again; the highest goes first

GRADING NOTE Only the cube roll gives everyone the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A fair lottery gives every participant the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

2609

QUESTION ID
S68-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what fair means here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Flipping one fair coin once is a fair way to choose one person from 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED

2610

QUESTION ID
S68-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A coin has only 2 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED

2610